AWS EBS (Elastic Block Store).

1. What is AWS EBS?

AWS EBS (Elastic Block Store) is a block-level storage service provided by Amazon Web Services that is used with Amazon EC2 instances.

Think of it like a virtual hard disk attached to a cloud server.

- It stores persistent data.
- Even if the EC2 instance stops or crashes, the data remains safe.
- Mainly used for databases, file systems, and applications that require high performance storage.

2. Key Characteristics

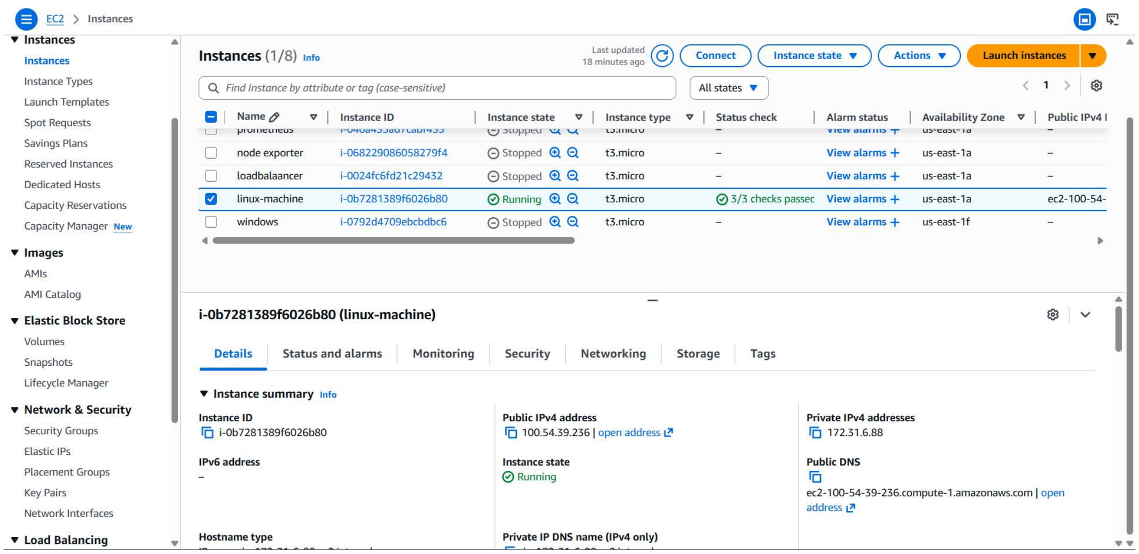
1. Block Storage
   Data is stored in blocks (similar to SSD/HDD).
2. Persistent Storage
   Data remains even after instance restart.
3. Attach/Detach Feature
   EBS volumes can be attached or detached from EC2 instances.
4. Snapshot Support
   Backup of EBS volumes can be stored in Amazon S3 as snapshots.
5. High Availability
   Replicated automatically inside the same Availability Zone.

3. Types of EBS Volumes

Volume Type	Description	Use Case
gp3 / gp2	General Purpose SSD	Web servers, small databases
io1 / io2	Provisioned IOPS SSD	High-performance databases
st1	Throughput Optimized HDD	Big data, analytics

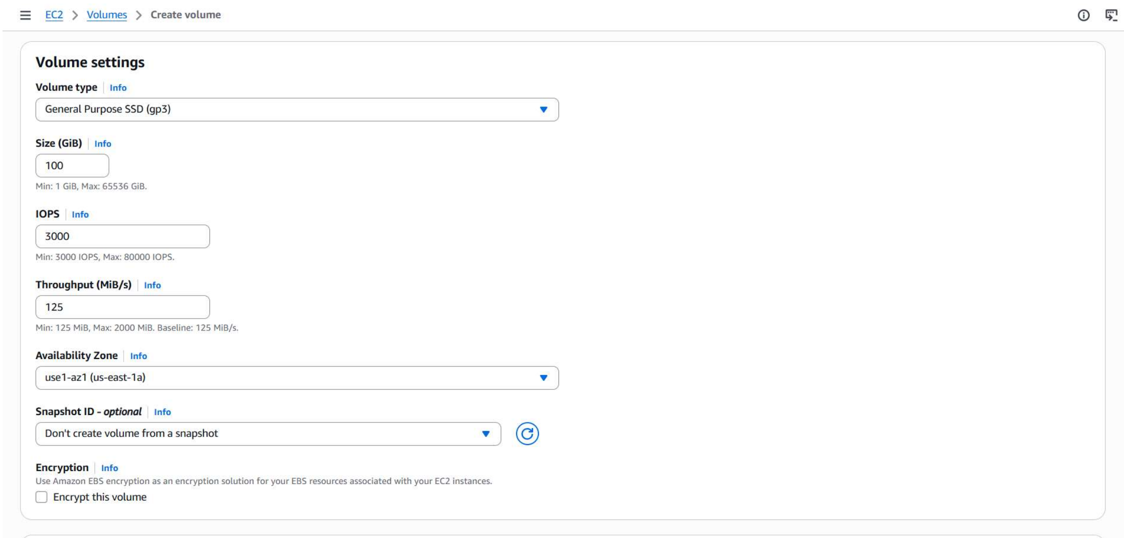
Volume Type	Description	Use Case
sc1	Cold HDD	Infrequent access

I have created an instance and then I have created an volume of 100 GB and attached to this instance.

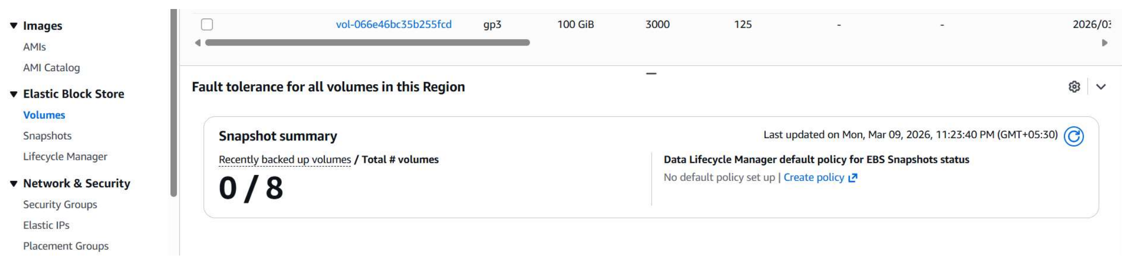


You can check using the **lsblk** command.

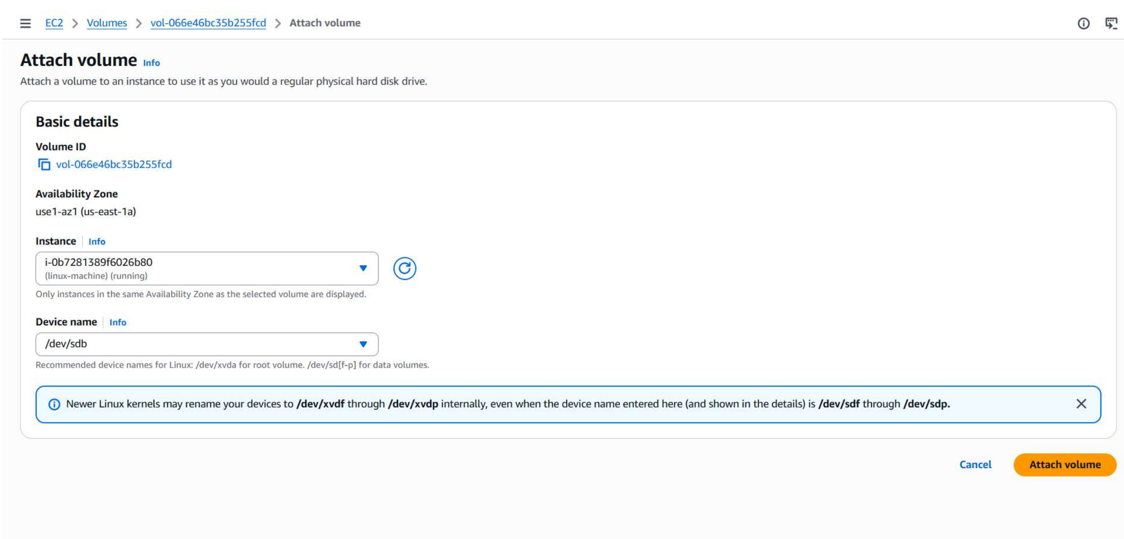
This is the created configuration storage of the new volume.



Created volume.



Then we have to attach to the instance but the **EBS** and **EC2** have to be created in same region and same AZ's. Then we have to select the running ec2 machine.

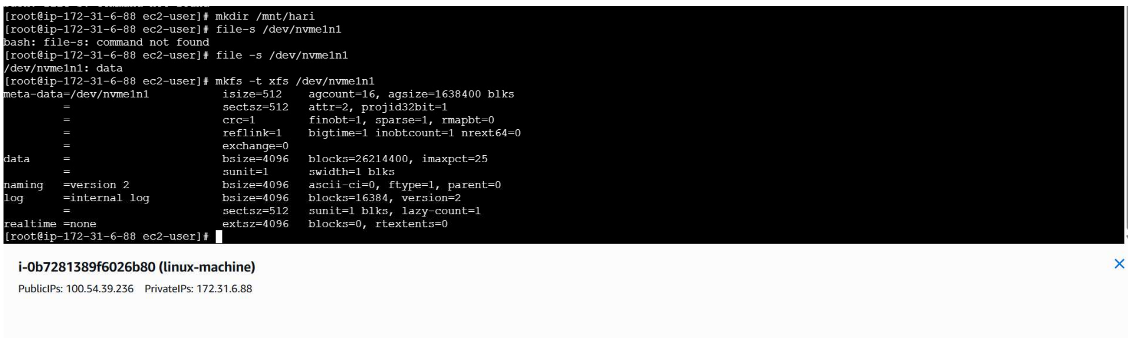


When we use **lsblk** we can see the un mounted volume with free space.

```
[ec2-user@ip-172-31-6-88 ~]$ sudo su
[root@ip-172-31-6-88 ec2-user]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0    8G  0 disk
├─nvme0n1p1   259:1    0    8G  0 part /
├─nvme0n1p127 259:2    0    1M  0 part
└─nvme0n1p128 259:3    0   10M  0 part /boot/efi
nvme1n1       259:4    0  100G  0 disk
[root@ip-172-31-6-88 ec2-user]# df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             459M     0  459M   0% /dev/shm
tmpfs             184M  440K  183M   1% /run
/dev/nvme0n1p1    8.0G  1.6G  6.4G  20% /
tmpfs             459M     0  459M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
[root@ip-172-31-6-88 ec2-user]#
```

file -s /dev/nvme1n1 >>>> Checks if the disk already has a filesystem.

mkfs -t xfs /dev/nvme1n1 >>>> Creates a filesystem on the disk.



```
[root@ip-172-31-6-88 ec2-user]# mkdir /mnt/hari
[root@ip-172-31-6-88 ec2-user]# file-s /dev/nvme1n1
bash: file-s: command not found
[root@ip-172-31-6-88 ec2-user]# file -s /dev/nvme1n1
/dev/nvme1n1: data
[root@ip-172-31-6-88 ec2-user]# mkfs -t xfs /dev/nvme1n1
meta-data=/dev/nvme1n1           isize=512    agcount=16, agsize=1638400 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=1 inobtcount=1 nrext64=0
         =                       exchange=0
data     =                       bsize=4096   blocks=26214400, imaxpct=25
         =                       sunit=1      swidth=1 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1, parent=0
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=1 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-6-88 ec2-user]#
```

i-0b7281389f6026b80 (linux-machine)
PublicIPs: 100.54.39.236 PrivateIPs: 172.31.6.88

mkdir /mnt/myvolume Creates a **mount point directory** where the volume will be attached.

mount /dev/nvme1n1 /mnt/myvolume Mounts the EBS volume to the directory. **df -h** Shows disk usage and mounted filesystems.

```
[root@ip-172-31-6-88 ec2-user]# mount /dev/nvme1n1 /mnt/hari
[root@ip-172-31-6-88 ec2-user]# df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             459M     0  459M   0% /dev/shm
tmpfs             184M  440K  183M   1% /run
/dev/nvme0n1p1    8.0G  1.6G  6.4G  20% /
tmpfs             459M     0  459M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
/dev/nvme1n1      100G  747M  100G   1% /mnt/hari
[root@ip-172-31-6-88 ec2-user]#
```

i-0b7281389f6026b80 (linux-machine)
PublicIPs: 100.54.39.236 PrivateIPs: 172.31.6.88